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Students' Beliefs About Agentic Engagement: A Phenomenological Study in Urban High School Physical Science and Engineering Classes

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Agentic engagement refers to students' proactive and constructive contribution into the flow of classroom instruction and activities. Given its potential as a student-initiated pathway to promoting motivation and learning, the goal of this study was to describe agentic engagement from the perspectives of racially diverse students in urban high school science classrooms. A secondary goal was to consider the relation between belonging to an underrepresented group in science (i.e., female, Black, and/or Latino) and thoughts and behaviors pertaining to agentic engagement. We conducted focus groups and interviews with 68 students using phenomenological methods to examine what, how, why, when, and who questions regarding the use of agentic engagement in science classes. Results focused on the following themes. (a) Students identified a variety of strategies for enacting agentic engagement (e.g., suggestions), though their use was described as rare. (b) Students emphasized the importance of style or approach for enacting agentic engagement in constructive and effective ways, with Black and Latino students noting the importance of avoiding disrespect and approaching teachers privately and girls mentioning using a group approach. (c) Students identified three rationales for agentically engaging, including to elicit teacher support, support personal experiences of motivation and learning, and create a more desirable learning environment for the classroom community. (d) Students discussed a variety of factors that made agentic engagement more or less likely, including teacher factors (e.g., encouragement, discouragement) and student factors (e.g., personality). Boys mentioned teacher encouragement more frequently than girls. Implications for theory, research, and practice are discussed.

Education Impact and Implications Statement

We conducted focus groups and interviews with 68 racially diverse students in urban high school science classrooms to better understand their experiences with trying to assert themselves in the classroom to influence the motivational quality of the instruction they receive (agentic engagement). Results revealed that students can identify a variety of strategies for agentic engagement but rarely use them or see peers use them. Nonetheless, they experienced agentic engagement strategies as effective, especially under the right conditions (e.g., teachers are encouraging, students assert agency with respect), for creating a motivating learning environment for themselves, their teachers, and peers in the classroom. Results highlight the potential of agentic engagement for transforming the motivational climate of the classroom and the need for teachers to be open to and supportive of it in their classrooms.

Keywords: agency, agentic engagement, motivation, qualitative, science

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Agentic engagement refers to students' proactive and constructive contribution to the flow of the instruction and activities that they receive (Reeve & Tseng, 2011). Students engage in a variety of behaviors, such as making suggestions, expressing their opinions, and asking questions, to influence their own learning and motivation in the classroom. Although it has a recent history, research to date has shown that the inclusion of agentic engagement within the larger engagement construct provides a fuller picture of students' involvement in classroom activities (Reeve et al., 2020). That is, agentic engagement predicts a wide-range of important motivation and achievement outcomes, over and above other forms of engagement (e.g., Patall et al., 2019; Reeve, 2013).

Building on the knowledge about the importance of agentic engagement as a student-initiated pathway to academic progress, the goal of this study was to describe the nature of this construct.

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More specifically, our purpose was to gain a deeper understanding of agentic engagement from the perspective of a racially diverse sample of students in urban U.S. schools to address questions about how they agentically engage and what beliefs they hold about this process. To this point, there is scant research that describes this construct from student perspectives. We documented findings from a qualitative inquiry to develop a framework for understanding agentic engagement from the perspective of high school students. Although agentic engagement is important across contexts, the increasing need for autonomy and independence during adolescence (Eccles et al., 1993; Erikson, 1968) makes understanding this phenomenon particularly important in the context of secondary school settings. In addition, we situated this study in physical science and engineering education to understand how this process occurs in context, particularly as it relates to female and racially minoritized individuals who have traditionally been marginalized in these environments. As such, a secondary goal of the study was to explore the relation between belonging to an underrepresented group in STEM (i.e., female, Black, and/or Latino) and their thoughts and behaviors as it pertains to agentic engagement.

Literature Review

Engagement refers to the extent to which a student is actively involved in a learning activity (Skinner et al., 2009). Traditionally, engagement was conceptualized as a multidimensional construct consisting of three distinct, yet interrelated pathways that led to positive learning outcomes (Christenson et al., 2012; Fredricks et al., 2004). Behavioral engagement describes students' involvement in terms of their effort, attention, and persistence; emotional engagement describes students' involvement in terms of the presence of interest, enjoyment, and other positive emotions; and cognitive engagement describes students' involvement in terms of their attempts to regulate their own thinking and learning processes (Sinatra et al., 2015). From the perspective of self-determination theory (Ryan & Deci, 2017), students experience engagement to the extent that it satisfies innate, psychological needs for autonomy (i.e., feeling that actions emanate from the self), competence (i.e., feeling successful or efficacious), and relatedness (i.e., feeling connected to and mutual caring with others). Research has widely documented that teachers who tap into these natural resources with supportive instructional practices enhance their students' engagement (e.g., Aelterman et al., 2019; Jang et al., 2016; Patall & Zambrano, 2019). As such, behavioral, emotional, and cognitive engagement can be thought of as the psychological-behavioral reactions to whether learning environments satisfy psychological needs.

As a fourth dimension of engagement, Reeve (2013) proposed agentic engagement to describe a form of engagement in which students proactively attempt to transform their learning experiences. That is, students do not just react intrapersonally to the learning environment, they also exert their agency by interactively enriching, personalizing, and modifying learning activities and instruction (Bandura, 2001). Like the other three types of engagement, agentic engagement also arises from the learning environment, but it is uniquely proactive and reciprocal. With behavioral, emotional, and cognitive engagement, a teacher may present a learning activity (e.g., a lab assignment) and students will react behaviorally and psychologically by showing varying levels of attention, interest, and strategic thinking. The reciprocal effect of students' behavioral, emotional, and cognitive engagement on the learning environment and other people in it is largely unintentional and inadvertent. In contrast, students who agentically engage go beyond a primarily intrapersonal reaction to the environment by engaging interactively and with initiative, for example, by speaking up in a constructive and proactive manner; they intentionally contribute to the learning environment (in addition to being influenced by it) in a reciprocal way (Reeve & Shin, 2020). Research suggests that agentic engagement is a statistically distinct construct that is related to the other three dimensions of engagement (i.e., behavioral, cognitive, emotional) but also independently predicts achievement and other motivational experiences, including psychological need satisfaction (e.g., Patall et al., 2019; Reeve, 2013). In the following subsections, we described how agentic engagement has been conceptualized, what the educational outcomes are, and the factors that influence its effectiveness.

Conceptualizations of Agentic Engagement

Agentic engagement involves action and behavior. It is defined as students' proactive attempts to learn and contribute into the flow of instruction and activities (Reeve & Tseng, 2011). Students who are agentically engaged are intentional and purposeful as they take actions, such as offering input and expressing their preferences. In taking these actions, students exert their desire to shape classroom activities and instruction so that they become more interesting and aligned with their own goals, values, and needs (Reeve & Tseng, 2011). Agentic engagement is also reciprocal in that students help create a pattern of ongoing interaction with the teacher that allows them to negotiate what is learned and how it is learned (Reeve, 2013). Beyond these features, the literature suggests that conceptualizing and operationalizing this construct is still developing (Montenegro, 2017). The actions and behaviors that typify agentic engagement have been informed by classroom observation and quantitative studies.

In creating the first agentic engagement measure, Reeve and Tseng (2011) relied on notes from extensive classroom observations as well as writings from motivational theorists (Bandura, 1997; deCharms, 1976; Ryan & Deci, 2000) to characterize how students exercise their agency in the classroom. Reeve (2013) subsequently modified the scale and operationally defined the agentic engagement construct with the following five items:

- I let my teacher know what I need and want.
- I let my teacher know what I am interested in.
- During this class, I express my preferences and opinions.
- During class, I ask questions to help me learn.
- When I need something in this class, I'll ask the teacher for it.

With these five actions, a scale was developed to capture the most common ways that students deliberately contribute to the flow of instruction.

Whereas the original scale by Reeve took into account the student (individually) and the teacher, Mameli and Passini (2019) proposed an extended scale to cover a larger variety of proactive student contributions. In particular, their scale was developed to

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account for student opposition in relation to their teacher and fellow peers. Using insights from classroom observations (e.g., Molinari & Mameli, 2013), Mameli and Passini added student questioning and criticism of the teacher's instructions (e.g., "If I do not agree with a teacher's statement, I tell him or her"), stance-taking among peers (i.e., "I defend my opinions even if they are not in line with those of my classmates"), and original contributions (i.e., "During class, it can happen that I introduce new issues or discussion topics"). This scale showed strong psychometric properties when tested with a sample of Italian high school students (Mameli & Passini, 2019).

Academic Outcomes of Agentic Engagement

Agentic engagement enables two key outcomes. First, students experience improved functioning in the form of learning, achievement, and other forms of motivation. For example, Reeve (2013) found that Korean high school students' early-semester agentic engagement predicted their end-of-semester achievement (i.e., grades), above and beyond students' other forms of engagement. Using a daily diary research design, Patall and colleagues (2019) also found that on days when U.S. high school students agentically engaged, they experienced increased autonomy and relatedness need satisfaction and increases in other forms of engagement. These studies are consistent with psychological theory which emphasizes the role of human agency in improving academic outcomes (Bandura, 2006).

Second, students indirectly experience improved learning circumstances in the form of enhanced teacher instruction and support. When students agentically engage, they help shape their teachers' instructional behavior and motivating practices. For example, in a study of Peruvian college students, Matos and colleagues (2018) found that students' early-semester agentic engagement predicted longitudinal increases in late-semester perceived autonomy-supportive teaching. Patall et al. (2019) also found that on days when students agentically engaged, they perceived increased teacher support for their autonomy on the same day, as well as over the course of an instructional unit. Perceived teacher autonomy support, in turn, predicted increased agentic engagement later in the instructional unit. Therefore, students' attempts to influence learning activities and instruction may recruit greater autonomy support from their teachers. These findings align with research demonstrating bidirectional relationships in which students' engagement and motivation could influence teachers' instruction and motivation support (e.g., Pelletier et al., 2002; Sameroff, 2009; Skinner & Belmont, 1993).

Factors That Affect Agentic Engagement

The literature suggests that the most important factor that could affect whether students enact agentic behaviors in the classroom as well as whether those initiatives are effective centers around a teacher's motivating style. According to Reeve and Shin (2020), a teacher's motivating style could range from one that is highly open to and respectful of students' perspectives and contributions (i.e., autonomy supportive), to a neutral style that is not concerned about students' perspectives (i.e., indifferent), to one that tries to force or pressure students to think, feel, or act in particular ways (i.e., controlling). How teachers deliver their instruction and respond to student initiatives in autonomy supportive, controlling, or indifferent ways could affect students' agentic engagement.

An autonomy supportive teaching style is open, understanding, and encouraging. The underlying idea of autonomy supportive instruction is consistently taking the perspective of students when structuring activities and communicating instruction in ways that allow students to feel they are engaging in those activities because they reflect their own values, preferences, interests, and goals (Aelterman et al., 2019; Reeve, 2009). Several core components have emerged as key to an autonomy-supportive teaching style, including providing opportunities for making choices, providing meaningful rationales that convey the usefulness and importance of tasks and topics, accepting students' negative emotions and empathizing, and being open to and incorporating students' perspectives by actively welcoming student input (Patall & Zambrano, 2019; Reeve, 2009). In addition to evidence from correlational-longitudinal studies (e.g., Matos et al., 2018; Patall et al., 2019), an experimental study has further corroborated agentic engagement as emerging from an autonomy supportive teaching context. Reeve et al. (2020) found that Korean physical education teachers who were trained to support their students' autonomy at the beginning of the year had students who reported increased agentic engagement and decreased agentic disengagement[1] (passivity) by midyear, compared with students whose teachers did not receive the training.

Whereas an autonomy supportive teaching style promotes students' agentic engagement, a controlling or indifferent teaching style is likely to suppress it (e.g., Assor et al., 2005; Rajala et al., 2016; Reeve & Shin, 2020). Controlling teaching practices are those that pressure students to think, act, or feel in particular ways, such as by using commands, punishments, and suppression. A teacher who adopts a controlling teaching style centers their own perspective as they make rigid and unilateral demands, using words like "must," "should," and "now," without rationales that consider students' perspectives, and resorts to the use of other forceful behaviors such as shaming and intimating students (Reeve, 2009). Patall, Steingut, Vasquez, et al. (2018) found that controlling teaching practices, such as the use of controlling messages and suppression of student perspectives, predicted increases in high school students' daily controlled motivation and disaffection in science classes.

A teaching style that is neither autonomy supportive nor controlling is one that is neutral or indifferent to students' perspectives. Teachers who are indifferent to students are unresponsive and show that they are disinterested in students' opinions, preferences, or suggestions in class (Bhavsar et al., 2019; Reeve & Shin, 2020). Although this teaching style is neutral, research suggests that neglecting to support autonomy has undesirable consequences for students' need satisfaction, engagement, and academic performance (e.g., Cheon et al., 2019).

[1] Because taking initiative is at the heart of what defines agentic engagement, and passivity is central to defining agentic disengagement, the two can be conceptualized as opposites. Empirical evidence suggests that the two constructs are negatively correlated (e.g., Jang et al., 2016). However, they are best conceptualized as two distinct constructs because they separate into distinct factors when analyzed using factor analysis, emerge from different antecedents, and predict different outcomes (Cheon et al., 2019; Jang et al., 2016; Reeve et al., 2020).

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Given that agentic engagement involves a student-teacher negotiation, power dynamics may also be a factor in influencing whether students enact their agency in the classroom (Reeve & Tseng, 2011; Winograd, 2002). In their interviews with Colombian students (aged 12–13), Pineda-Báez and colleagues (2019) found that teachers who conveyed authority and power discouraged students' agentic engagement. In addition, they reported that peers who tease and bully could also be detractors.

Beyond these teacher and student factors, the literature suggests that students who agentically engage may do so because of an underlying belief that such behavior will actually change their learning experiences. In a randomized controlled trial, Patall and colleagues (2021) randomly assigned undergraduate college students in chemistry and physics courses to a web-based agentic engagement orientation intervention or one of two control conditions. They found that students who received the target (agentic) intervention reported a more agentic mindset (an orientation toward viewing their motivation and instructor practice as malleable and responsive to agentic behaviors), which in turn, predicted greater student-reported in-class engagement, instructor autonomy support, and need satisfaction. These findings suggest that understanding student beliefs about agentic engagement may be critical in supporting student motivational processes.

Gender, Race, and Agency in the High School Science Classroom

In the United States, Black and Latino/Hispanic students are underrepresented in science, technology, engineering, and mathematics (STEM) education and careers, whereas female students are particularly underrepresented in the physical sciences and engineering (National Science Board, 2020). Undoubtedly, the classroom experiences and beliefs of girls and students of color during high school play a critical role in predicting whether they enroll and persist in STEM during college and for their careers (Tai et al., 2006). Students of color and female students often feel disconnected from science and report a lower sense of belonging, owing in large part to sociocultural systems that privilege majority groups (usually White males; e.g., Cheryan et al., 2017; Rainey et al., 2018). Given disproportionate levels of participation among female, Black, and Latino students in STEM education, it is important to consider how agentic engagement behaviors and experiences of these students may be different from their counterparts.

First, although students may want to voice their opinions and express themselves in class, some female, Black, or Latino students may be reluctant to enact those behaviors due to worry about confirming negative societal and science-specific stereotypes about themselves or their social identity group (Crocker & Major, 1989; Steele et al., 2002). Second, it may be possible that the act of asserting one's agency is not compatible with female, Black, or Latino students' cultural or familial values and goals that emphasize communion and interdependence over agency and individualism (Boucher et al., 2017; Jackson et al., 2016; Tyler et al., 2006; Zambrano et al., 2020). Third, it may also be possible that teachers hold biases of female, Black, or Latino students in physical science and engineering, and may inadvertently fail to respond supportively or may ignore or suppress students' agentic contributions. For example, Patall, Steingut, Freeman, et al. (2018) found that, compared with male students, female high school students experienced less daily autonomy support from teachers in physics, chemistry, and engineering courses. In turn, these differences explained girls' lower daily engagement compared with boys. Relatedly, it is also possible that agentic behaviors such as expressing disagreements with teachers' perspectives may be misinterpreted by teachers as "disrespect" or "defiance" when voiced by Black or Latino students (Gregory et al., 2010; Okonofua et al., 2016). Taken together, any effort to understand the motivational experiences of students in physical science and engineering, especially those that have consequences for (dis)empowerment, as student agency does, should be considered in light of group differences and historical and contemporary inequities (Usher, 2018).

The Present Study

By and large, the actions and behaviors that typify agentic engagement, as well as the academic consequences and factors that influence its effectiveness, have been informed by classroom observation and quantitative studies. What is missing from the literature are student voices. A qualitative inquiry of how students understand and experience agentic engagement from their own perspectives may yield new insights. We conducted the present study to better understand the phenomenon of agentic engagement. Our main goals were to describe how students agentically engage in the urban high school physical science classrooms, if at all, and what beliefs they hold about this process. We focused on three aspects of student beliefs about agentic engagement, including how, why, and when they agentically engage. A secondary goal of the study was to consider the relation between belonging to an underrepresented group in STEM and their beliefs and behaviors as it pertains to agentic engagement. Finally, we sought to juxtapose our description of the findings with the current state of the literature and discuss implications for teacher practice and future research.

Method

Methodological Framework

The goal of this study was to describe the nature of agentic engagement as experienced by a diverse group of students in urban high school science classrooms. We used phenomenology because agency is a relatively new dimension of engagement. Phenomenological design is appropriate when the goal is to explore the meaning of some phenomenon, in relation to lived experiences of people; it is especially useful when working with a phenomenon about which little has been written. This type of research describes and interprets the perspectives of the participants in terms of "what" individuals experience and "how" they experience it (Wertz et al., 2011). Therefore, phenomenological data are, by nature, descriptive—the data are not appropriate for constructing abstract theories or models, testing causal hypotheses, measuring psychological constructs, or assessing quantitative relationships. Rather, the results are intended to provide a deep understanding of a phenomenon as experienced by a group of people from which testable hypotheses can be generated (Flowerday & Schraw, 2000).

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Phenomenology was first introduced by Edmund Husserl (1931) as the philosophical study of the structures of consciousness as experienced in the first-person point of view. As such, this theoretical and methodological lens emphasizes intentionality of consciousness; that is, the directedness of experience toward aspects in the world. Phenomenological procedures are by nature broad, because they involve answering the basic question "what is it" while putting aside prior understandings of the phenomenon that would otherwise narrow the study. In thinking about how we wanted to approach data collection and analysis, we searched for general analytic procedures that were explicit and systematic and would allow us to describe agentic engagement so that it can be understood in a deep, holistic, and comprehensive way. We found that Giorgi's modified Husserlian approach (Giorgi, 1985, 2009), which followed Husserl's tradition but adapted it from philosophy to psychology, met these needs. This approach involves assuming the phenomenological attitude, or "bracketing" one's preconceptions and preunderstandings of the subject matter, collecting concrete, real-world examples from people who have experienced the psychological phenomenon, such as through interviews, and analyzing their meanings and structures (Wertz et al., 2011). Giorgi articulated several essential steps for phenomenological data collection and analysis, including: (a) reading with an open attitude for a sense of the whole; (b) differentiating data into meaning units; (c) reflecting on the psychological significance of each meaning unit; and (d) clarifying the psychological structure(s) of the phenomenon. Then, the findings are presented as a description of the phenomenon by integrating researcher insights with participants' own words (Giorgi, 2009; Husserl, 1999; Leedy & Ormrod, 2013; Wertz et al., 2011). We describe these steps in more detail in the following sections.

Participants

Data were collected from seven public high schools located in a large, urban county in the United States. School characteristics and demographics are shown in Table 1. To be eligible to participate, students had to be currently enrolled in a physical science class. We collaborated with 14 physics, chemistry, environmental science, and engineering teachers who had a combined total of 52 relevant class sections in our effort to make students aware of the opportunity to volunteer to participate in the study. The sample size and criteria for selecting the students were made a priori, with some constraints due to availability. From the list of 129 students who had signed up and agreed to participate, we selected students based on gender and race to hear from diverse student perspectives. Given the demographics of the schools we worked in, our goal was to collect data from six to 12 participants from each gender (male, female) by race/ethnic (Black, Latino, White/Asian) subgroup. Note that Asian and White students were combined; this grouping was based on how people are typically stereotyped in science (i.e., Asian and White students share a mostly positive stereotype). We selected from the list of students to form race by gender focus groups; students were invited for individual interviews when groups could not be formed due to logistical reasons, such as availability or scheduling.

Sixty-nine students participated in either a focus group or an interview. Data from one student interview was removed from the study due to audio-recorder malfunction. Of the 68 remaining participants, 41 (60%) identified as girls, 26 (38%) as boys, and one (1%) as nonbinary; racially, 23 (34%) identified as Latino/Hispanic, 15 (22%) as Black/African American, 14 (21%) as Asian, eight (12%) as White, and eight (12%) as multiracial; 11 (16%) were freshman, 14 (21%) were sophomores, 26 (38%) were juniors, 15 (22%) were seniors, and two (3%) did not report their class level or their age. Participants ranged from 14 to 18 years old; the median age was 16. It is important to note that the racial groups are diverse; for example, some White students in this sample (e.g., Armenian) may not be entirely representative of what other research samples refer to as White students (e.g., European Americans). Information about the diversity of racial categories is shown in Table 2. It is also important to note when interpreting findings, that Latino and Black student participants were more likely to come from less affluent schools, who had fewer White students enrolled. In contrast, White and, to some extent, Asian student participants came from somewhat noticeably more affluent schools in which fewer Black and Latino students were enrolled.

Data Collection

To understand how urban high school students experience agentic engagement, we collected data through conversational semistructured focus groups and individual interviews. Although we recognize that traditionally most phenomenological studies involve one-on-one interviews, because focus on the essence of a phenomenon requires individuals to describe their personal experiences, we agree with phenomenologists who argue that focus groups are congruent with phenomenological research, and further, that they are actually advantageous. Focus groups stimulate discussion, open up new perspectives, and allow participants to

Table 1
Characteristics of Schools

School	Type	Size	Latino	Asian	Black	White	Other	Economically disadvantaged
Palm	Charter	600	98.5%	0%	0.8%	0.5%	0.2%	96.3%
Stanton	Charter	500	84.9%	0.4%	11.9%	1.5%	1.3%	95.7%
Gilbert	Charter	600	86.8%	0%	12.7%	0.2%	0.4%	92.9%
Vista	Magnet	1,000	56.0%	32.7%	6.4%	4.4%	0.5%	78.6%
Carnegie	Magnet	2,400	57.8%	22.6%	2.4%	16.3%	0.9%	73.3%
Suncoast	Public	1,500	54.0%	18.1%	19.5%	3.1%	5.3%	70.5%
Fairview	Public	1,500	25.9%	37.6%	1.7%	27.0%	7.6%	20.3%

Note. School names are pseudonyms. Sizes are rounded to the nearest hundred to preserve the anonymity of the schools. The economically disadvantaged percentage was measured using the percentage of students receiving free or reduced-priced lunches.

Table 2
Participant Profiles by Session

FG/I	*n*	School	Gender	Age(s)	Race group	Specific racial/ethnic information
FG	6	Carnegie	F	16–18	Asian/White	Chinese; Korean; Armenian; Armenian/Lebanese; Russian; Jewish (White)/Latina
FG	4	Fairview	F	15–17	Asian/White	White (European); White; White (European)/Japanese; Chinese/ Japanese/ Korean
I	1	Vista	F	15	Asian	Korean
I	1	Vista	F	16	Asian/White	Chinese/Irish
I	1	Carnegie	F	17	Asian	Filipino
I	1	Vista	F	15	Asian	Cambodian/Chinese
I	1	Fairview	F	15	White/Asian	Norwegian/Hawaiian[a]
FG	3	Carnegie	F	16	Latina	Salvadoran; Mexican; Salvadoran
FG	4	Palm	F/NB	14	Latina/x	Mexican; Mexican; Chicana; Mexican
FG	3	Suncoast	F	14–16	Latina	Peruvian; Mexican/White; Mexican
FG	4	Gilbert	F	16–17	Latina	Nicaraguan; Latina; Mexican; Mexican
I	1	Stanton	F	17	Black	African American
I	1	Stanton	F	16	Black	African American
FG	7	Gilbert	F	17–18*	Black	African American; Black; Black; Black; Black; Black; Black
FG	4	Gilbert	F	14–16	Black/Latina	Black/Latina; Black; Black; Black
I	1	Carnegie	M	16	Asian	Filipino
FG	2	Carnegie	M	16–17	Asian	Bengali; Filipino
FG	4	Fairview	M	15–17	Asian/White	Taiwanese; Vietnamese/German; Chinese; Chinese
I	1	Vista	M	15	Asian	Chinese
I	1	Vista	M	15	Asian	Chinese
I	1	Fairview	M	15	White	Caucasian
I	1	Carnegie	M	17	White	Irish American
I	1	Suncoast	M	17	Latino	Latino
FG	4	Carnegie	M	16–17	Latino	Salvadoran, Mexican, Mexican, Mexican
FG	4	Vista	M	14–16	Latino	Nicaraguan, Mexican, Salvadoran, Latino
I	1	Vista	M	17	Latino	Mexican
I	1	Suncoast	M	14	Black/White	Black/White
I	1	Suncoast	M	16	Black/Latino	African American/Mexican
I	1	Suncoast	M	15	Black	Black
I	1	Gilbert	M	18	Black	Black
I	1	Suncoast	M	16	Black	Black

Note. FG = focus group; I = interview; *n* = sample size; F = female; M = male; NB = nonbinary. Asterisks denote missing participant age information. For specific ethnic information, some students did not provide details; for these cases, we included the only information they provided (e.g., White).
[a] One student who identified as Norwegian and Hawaiian was grouped with the White/Asian group's responses given that no other students who identified as Pacific Islander were included in our sample. Although some bodies of research combine Asian American and Pacific Islanders into one racial category, we do note that differences exist (Hall, 2015).

elaborate their views in response to encouragement or defend them when challenged by other group members (Bradbury-Jones et al., 2009; Wilkinson, 1998; Wilson, 2007). In addition to these benefits, our decision to conduct focus groups was further informed by the topic of inquiry (i.e., agentic engagement is a new term that students may not be used to talking about) and our interest in considering the experiences of marginalized students in science (i.e., affinity groups may draw out whether agentic engagement manifests in relation to gender- and racially specific lived experiences of girls and people of color). Focus group moderators encouraged all participants to voice their thoughts and experiences.

Focus groups were 1.5-hour sessions and interviews ranged from 30 to 45 minutes. To maximize variation and hear from diverse student perspectives, we strategically sampled students based on gender and race or ethnicity. Because girls (compared with boys) and Black and Latino students (compared with Asians and Whites) face negative stereotypes about their ability and belonging in science and engineering (e.g., Murphy et al., 2007; Schmader & Hall, 2014), we used gender and race/ethnicity characteristics to create relatively homogeneous groups that would minimize feelings of threat, including the gender and race/ethnicity characteristics of the interviewer/moderator (e.g., Black female groups were facilitated by a Black female researcher; Latino male groups were facilitated by either a Black/Latino male researcher or a Latina female researcher). Six interviewers/moderators were involved in collecting the data; all underwent training in interview and focus group data collection techniques and use of the protocol.

All focus groups and most interviews were conducted in-person at or near the students' school, inside of a classroom or at outside seating areas (teachers were not present during data collection). Eight interviews were conducted online because students could not meet in person. Students who attended focus group sessions created name tags (because they came from different classrooms), introduced themselves to the group, and were provided food and beverages; those who attended in-person interviews were also offered food and beverages. Graduate research assistants conducted the interviews and focus groups.

The interviews and focus-groups followed the same protocol and set of semistructured questions (see Appendix A). First, research assistants introduced themselves and the study to student participants. Students were told that there were no right or wrong answers, and that we were simply interested in hearing their perspectives. Next, they were presented with a diagram (see Appendix B). The diagram was a model adapted from Reeve (2013), which presents how teacher support for student motivation (e.g., creating interesting activities)

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influences the four types of engagement (emotional, cognitive, behavioral, and agentic), which then influence student interest, achievement, and other academic outcomes. After examining the figure, participants were asked what they thought the diagram was trying to describe; students were invited to agree or disagree with ideas as presented in the figure or expressed by the researcher (or other students). Then, the interviewer/moderator provided students with a basic conceptual description of engagement types and the possible relationships between engagement and the classroom environment, merely as a way to introduce the topic of discussion.

After the description, students were asked three overarching, open-ended discussion questions and throughout the discussion researchers adapted to students' narrative logic. That is, the main questions focused broadly on students' beliefs (e.g., "What do you think of this idea that students can change their own motivation in physical science class with the way they engage in class?") as an entry point to student perspectives; and then, researchers followed with questions based on students' responses (e.g., Have you done something like that in your class before? Why did you do that? What was that like?). Therefore, the semistructured interview questions were loosely designed to go from broad (beliefs) to specific (feelings and experiences) to allow the discussion to capture a wide range of thoughts, feelings, experiences, beliefs, and ideas related to agentic engagement. Moreover, by asking broadly about what students think of this idea first, our aim was to reduce the researchers' influence (i.e., avoiding a structured discourse based on the researcher's own logic of thought) by allowing students to formulate their own beliefs and then allow them to tell us what that looks like for them. Students were compensated with a $25 gift card.

Individual interviews were conducted in six of the seven schools, and focus groups were conducted in six of the seven schools. In total, we conducted 19 interviews and 12 focus groups (31 sessions). These sessions were audio recorded and transcribed. Focus groups were also video recorded to keep track of responses from each participant during transcription. The interview and focus group session profiles are presented in Table 2.

Data Analysis

We analyzed the full data set by using systematic procedures. First, two researchers (J.Z., E.A.P.) read through several transcripts, without taking notes and without an agenda, to get a sense of the data. This first step is important for phenomenology because it allows analysts to be open to participant perspectives (Giorgi, 2009; Posselt, 2018). Second, the two researchers inductively analyzed all of the transcripts by identifying significant statements. Using the significant statements, they each created drafts for a codebook with narrow meaning units (or codes) organized under broader meaning units (or themes; Wertz et al., 2011). The two analysts then came together to reflect and discuss their initial drafts and worked collaboratively to create one codebook. Third, the codebook was tested; five authors and other members of the research team independently coded one focus group transcript. We discussed discrepancies in our codes and our understanding of the meaning units and revised the codebook accordingly. This process was repeated with another focus group transcript and a smaller group of five analysts. Fourth, three analysts (J.Z., A.A.U.K., C.A.) subjected the entire data set to the full set of codes as follows: each analyst independently engaged in line-by-line coding of the transcript; the codes from each analyst were compiled into a single file to compare the codes; the three analysts resolved discrepancies through deep analysis of the data and discussion, including reflection, self-questioning, and questioning each other. This process was repeated for all transcripts.

Fifth, the resolved codes were transferred to NVivo for organization and further examination (e.g., frequency of coded units). Sixth, one analyst (J.Z.) reflected on the list of codes, their meanings, and how they were organized as a structured whole of the phenomenon (Charmaz & McMullen, 2011; Giorgi, 2009). At this stage, some meaning units were reorganized (e.g., building relationship with teacher code was moved from meaning unit agentic behaviors to meaning unit agentic style or approach), and themes were further disaggregated and aggregated into clusters. As a result of this analysis, themes were renamed to capture new meanings. Seventh, we considered whether some meaning units were more prevalent or salient for particular groups of people. This was done by going through each code and viewing how many students from each gender by race subgroup mentioned those comments. When some pattern was detected, we reflected on what it might mean for participants with a certain social identity to have those beliefs, experiences, or concerns in a high school physical science context. We reported that no patterns by race or gender were identified when students from all or most backgrounds were making similar comments. Eighth, as the last stage of phenomenological analysis, we presented the findings as an integration of our understanding with the participants' words. The findings were then juxtaposed with the current literature that exists on agentic engagement.

Trustworthiness and Rigor

Our goal was to produce trustworthy and rigorous research. To meet our goal, we employed well-established strategies from qualitative research, broadly, and phenomenological research, specifically. These included the use of multiple forms of data collection (i.e., interviews and focus groups), diverse participants (see Table 2), diverse interviewers and moderators (i.e., two Black, two White, one Latina, one Asian), transparency of the analytic process, coder agreement checks (i.e., resolving discrepancies), and use of several analysts at multiple phases of data analysis. This last component was important to the way we approached this phenomenological study. Analysts regularly articulated their interpretations of the data and other analysts interrogated and built on those interpretations. We employed multiple analysts in a systematic way (steps 1–5 in the prior section) to slow down the process of understanding, to function as a check on each analyst's judgements, assumptions, and preconceptions, and to facilitate reflection of the meaning units—all of which are essential for rigorous and trustworthy phenomenological research (Wertz et al., 2011).

Results

The findings are organized around three broad themes that describe student beliefs about agentic engagement (see Figure 1 for an outline). Strategies for agentic engagement refer to the methods that students used to proactively and constructively contribute to their learning and instruction. Rationales for agentic engagement refer to the reasons why students chose to use strategies for agentic engagement. Finally, criteria for agentic engagement refer to students' beliefs of the factors that affected the use

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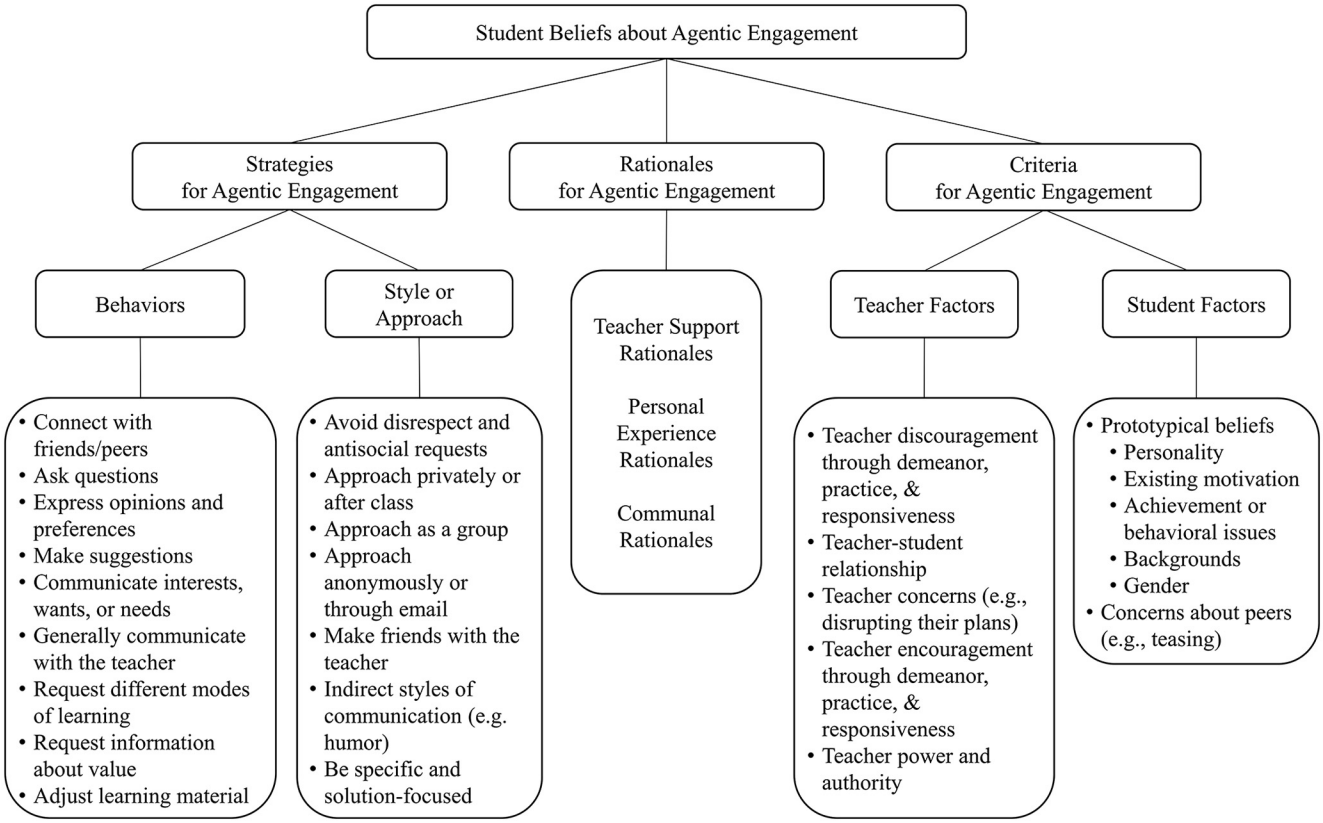
Figure 1
Student Beliefs About Agentic Engagement



and effectiveness of agentic engagement strategies. There was some correspondence between the questions asked of participants and the themes; however, responses to all questions (i.e., all transcript data) were used to inform the themes. We describe these themes in the next sections, including areas where findings were more salient among some groups of students relative to others. Refer to Table 3 for the number and proportion of students who made each type of comment. Practical and theoretical consequences of our findings, as well as their relationship to existing educational psychology research, are reserved for the discussion section.

Strategies for Agentic Engagement

Students were asked to describe what they can do to change their motivation and learning experiences in the classroom. They responded with a variety of strategies for agentic engagement. The strategies were divided into two subthemes: agentic behaviors and agentic approach or style.

Agentic behaviors refer to the self-initiated actions that students took in the classroom to proactively influence their learning and motivation. Two students described behaviors that depicted high levels of agency in which they took a new action out of their own volition. Esteban's description of a time when he initiated a science activity was one such example:

> One time, [my teacher] was telling me a story of how his microwave—last year, I think so—like exploded, because some students put ramen without water, and it just blew up. So I asked him, "Can I make a model?" after. And he is like, "Yeah, sure." So I did, and I gave it to him. So he gave me extra credit. [Esteban, Latino, male, age 16]

The more common pattern in the data was for students to take some action in relation to the teacher to adjust, calibrate, or enhance learning instruction and activities. These behaviors included asking questions, expressing opinions and preferences, making suggestions, and communicating interests, wants, or needs to the teacher. Miguel provided us with an example of how a student suggestion was used to enrich a learning activity by making it more personally relevant in his class:

> We were studying urban development in environmental science class. And initially, it was just going to be focused on the topic of a specific area, whichever one you wanted. But one kid suggested that we could do areas that we live in, and see how urban development was affecting that area. So that just made us—that made us more engaged, and it made us feel more attached, and our teacher was open to that idea. [Miguel, Latino, male, age 17]

In addition to these broad categories, we also identified three, more specific, behaviors. These included asking for different modes of learning, such as requesting to do "a minilab for this topic to get a better grasp of it;" requesting information about value, such as asking "What's the purpose of this?" and asking for the material to be adjusted, such as telling the teacher when

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Table 3
Number and Proportion of Students per Code

Theme	Code	No.	%
Strategies			
Behaviors	Connect with peers	48	71%
	Ask questions	46	68%
	Express opinions and preferences, make suggestions, communicate interests, wants, or needs	43	63%
	General communication with teacher	22	32%
	Request different modes of learning	12	18%
	Request value information	7	10%
	Adjust material	6	9%
Style or approach	Avoid disrespect	20	29%
	Approach privately or before/after class	17	25%
	Group approach	10	15%
	Anonymous or email	8	12%
	Make friends with teacher	8	12%
	Indirect styles	7	10%
	Specific/solution-focused	3	4%
Rationales	Teacher support	51	75%
	Personal experience	27	40%
	Communal	18	26%
Criteria			
Teacher-factors	Discouragement	25	37%
	Teacher-student relationship	17	25%
	Teacher concerns	17	25%
	Encouragement	13	19%
	Power and authority	12	18%
Student-factors	Prototypical beliefs		
	Personality	18	26%
	Existing motivation	11	16%
	Achievement/behavioral issues	11	16%
	Backgrounds	5	7%
	Gender	2	3%
	Peer concerns	16	24%

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something was too hard or too easy. We also identified more general statements related to communicating with the teacher, but these statements lacked specificity (e.g., "Talking to the teacher, in general, makes everything better"). From the students' perspectives, enacting agentic engagement behaviors meant participating in a way that influences or nudges the teacher toward better supporting their learning and interests in the classroom.

Seventy-one percent of students also mentioned the importance of connecting with friends or peers in the classroom. They discussed the need to "surround yourself with the right type of people" to "become motivated," initiate group work to make activities more fun, ask peers for explanations when they did not understand the teacher, and help their peers to "set a good environment for yourself" and everyone else. Chris described this strategy as follows:

> Like [another student] said, trying to stay motivated, and with that, help everyone around you. Give off your energy to everyone around you and make them want to—like, if you are working with a lot of people who don't talk, start off the conversation: Ask "What's your name?" Just ask about them, and maybe they will give in and start talking. [Chris, Latino, male, age 16]

Student descriptions of connecting with friends or peers involved taking action to solicit support from classmates and also being the person that helps classmates and initiates collaboration to cultivate a more positive and engaging environment. Students felt that connecting with their peers and cultivating a "sense of community" was important for their learning; they recognized that their peers were resources that could significantly impact their engagement in learning activities. We discuss the related idea of communal rationales in the next section. Overall, we did not detect any patterns by race and/or gender for the agentic behaviors.

Next, agentic approach or style refers to the actions that students took to create more favorable conditions or situations so that agentic engagement behaviors were effective when enacted. These strategies were almost exclusively discussed as ways to approach the teacher with their questions, requests, or suggestions so that their bids were more likely to be successful. Common strategies for enacting agency in the classroom were to approach the teacher privately or before or after class to avoid upsetting the teacher, or to facilitate greater understanding of the students' needs, and to avoid disrespect and antisocial requests, such as not telling the teacher "I kind of find this boring," or asking the teacher for "the formula for how to make a bomb." Other agentic approaches or styles that students used to solicit support from their teachers included: Approaching the teacher as a group to show urgency for their requests and persuade the teacher because there is power in numbers; approaching anonymously (e.g., leaving a note) or through e-mail, particularly for students who may not feel

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comfortable, or for students who want to avoid their teachers' negative reactions; making friends with the teacher or telling the teacher about yourself personally because they "start giving you some benefits;" and using indirect styles of communication, such as "nodding really obviously," facial expressions, and using humor (e.g., making jokes) to make the teacher "more happy to teach."

There were some agentic engagement approaches or styles that were more salient among particular groups of students compared with others. Black and Latino students accounted for 90% of those who commented on the importance of avoiding disrespect and antisocial language, and 88% of those who commented on approaching the teacher privately or outside of class time (across gender). Conversely, all comments that described making friends with the teacher were made by Asian and White students (across gender). Girls (across race) made all comments about approaching the teacher as a group. In addition, there was another agentic approach that was mentioned in only one focus group session (Latina girls), but it was a salient topic in their discussion: agentic engagement strategies should be specific or solution focused. Karina described why she felt being specific was important:

> I feel like, instead of just being general, like you have to be kind of specific with it. I feel like that also does definitely make it more of like an, "Oh this is what I want to change," instead of like, "Oh I don't like this," and that is because if you give a suggestion on how they can fix it then that's mutually beneficial because you can help [teachers] help you, because if you—say a class is boring, the teacher doesn't know what part of it you find boring, or what to do. Maybe they will cut off a part that was actually interesting you or intriguing you. [Karina, Latina, female, age 16]

Our understanding of the agentic style or approach finding was that it demonstrated the extent to which students were intentional, strategic, and diplomatic in maximizing the effectiveness of agentic behaviors. Students believed that the way they approached their teachers could significantly influence the extent to which their suggestions or questions were understood, considered, and integrated into learning instruction and activities. They felt that to be successful, they should be respectful and strategic. Moreover, they believed that students who are not respectful or are not being strategic may not be agentically engaging at all; instead, their actions might be considered complaints.

Put together, students not only discussed behaviors that they used to proactively contribute to their own learning and motivation, they also discussed constructive ways to enact those behaviors so that they were more effective. However, although students appeared to understand our questions about agentic engagement and discussed their beliefs about agentic behaviors and approaches, we found that 10 students said that they themselves or their classmates do not behave in this way often, if at all. More students said that they "don't feel like anyone really expresses themselves in class to make the teacher better understand or teach better—it's just not something that I see happen" [Samuel, Black, male, age 18] compared with four students who said that they do agentically engage in class often and see others do so as well. We discuss the factors that influence student use of agentic strategies later in the criteria for agentic engagement section of our findings.

Rationales for Agentic Engagement

When students discussed agentic engagement strategies, they often provided a reason for behaving in that way; if they did not, we probed their responses to understand their intentions. Students provided different reasons regarding why and for what end they used agentic engagement strategies. Rationales were organized into three subthemes: teacher support rationales, personal experience rationales, and communal rationales. These rationales describe what students hoped to achieve when they agentically engaged in the classroom.

The most frequently cited rationale for agentic engagement was teacher support. Teacher support rationales reflected the belief that students could use agentic strategies, such as asking questions and making suggestions, to indirectly influence the teacher to better support their own learning and motivation. When students agentically engaged, they sought to improve instruction of class topics and move the teacher toward more interesting, engaging, and personally relevant learning activities. Students often described this process as using agentic engagement strategies to help their teachers better help them. One student also described the agentic engagement process as "positive reinforcement" for their teachers, such that "when [teachers] can see that the students are engaged and they enjoy [the activity], they'll be more likely to do that again" [Miguel, Latino, male, age 17]. The following statement from Jacob is an example:

> I think [students] can help [teachers] because if a teacher is teaching a student about this science topic, and the students don't like it that way, the teacher will change it because the students are giving feedback . . . So if I can give an example, I remember the teacher, he gave us these electricity kits, and we actually really liked them, but he thought it was failing because when he gave it to his other students, his seniors, the seniors didn't really love them . . . . So he thought, "Hey, I don't think this is a great way to do it . . ." But we're like, 'we' us freshmen, we're like, "Yes, this is actually really interesting. This is fun, we get to learn about electricity, about the neutrons, and about everything." We thought it was fun, so he continued to do it. [Jacob, Black/White, male, age 14]

The data suggest that students understood agentic strategies as useful in guiding the teacher's instruction based on their own needs and preferences to enhance their understanding and interest in class topics and activities. There were no notable differences by race and/or gender for teacher support rationales.

In contrast to indirect benefits through the teacher, students also mentioned personal experience rationales. These rationales refer to the way students believed agentic engagement strategies could directly enhance their own experience of learning and motivation in the classroom. Some students described feeling "better about yourself" when using agentic engagement strategies. Two examples are as follows:

> I think it feels good because you are able to explore what you want to explore . . . you are basically motivating yourself to do something that you want to do. [Mateo, Latino, male, age 15]

> Because you could be open in class, it makes the experience better . . . So, when you communicate with teachers it makes you want to learn. [Nia, Black, female, age 16]

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As these quotes demonstrate, students discussed positive experiences and motivational benefits when they were open and expressed themselves in class. Although rationales that described direct benefits to the student were the second most stated rationale, we found that these descriptions were shallow and subtle: students did not elaborate as much about this type of rationale in comparison to the other two rationales. We did not detect any patterns by race and/or gender for these rationales.

Next, some students described communal rationales. These students clearly articulated the idea that agentic engagement strategies do more than improve their own learning experiences; they also help create a more motivating and "more fun and energetic" learning environment for everyone else in their class, including their peers and teachers. They expressed the idea that "it just takes one person" or a small number of students to transform the learning environment for everyone in the classroom. For example, "if you do not have those people, like those three or four people that speak out all the time, then nobody is interested in class anymore" [Aiko, White/Asian, female, age 17]. Students believed that agentic engagement was socially influenced. For example, one student said "I assumed that maybe the other students would be more comfortable to bring their own stuff because they saw me do it" [Tony, Latino, male, 14]. The following is a comment from Sandra that depicted a communal rationale for agentically engaging in the classroom:

> Asking the simple question when they're confused, or making a suggestion in class, like "Oh, can you explain that again?" I feel like by that simple initiation, you get a different environment because everyone feels like they are confident to speak up, and actually pay attention, because they've noticed that they are not the only one who's alone and feeling confused. [Sandra, Latina, female, age 16]

Students described communal rationales for agentic engagement almost in terms of a virtuous cycle, in the sense that they should express themselves and "encourage others to speak up" so that "everybody benefits." In addition, our impression was that students thought of agentic engagement as generating energy in the classroom, whereas a lack of agentic engagement in the classroom (i.e., passivity) would contribute to a low-energy or negative classroom climate. We did not detect any patterns by race and/or gender when examining the number of students who made communal rationale comments. However, we do note that Latino students who made communal rationale comments were more vocal about them[2] (59% of these comments were made by Latino students).

In general, students believed that using agentic strategies could help create a more comfortable, open, and engaging learning environment for themselves, their teachers, and their fellow classmates. Although most of students' attempts to influence instruction and activities operated within the bounds of their immediate classroom learning environments, this was not always the case: two students discussed using agentic strategies to move to a different learning environment altogether (i.e., "change the teacher, or ask for a replacement"). Lisa explains how and why she did this:

> I had a teacher—she was being rude to me; she was being racist . . . I was speaking in a presentation, and I misunderstood this one word, and the teacher I heard say, "Can Mexicans talk right?" And I thought that was pretty rude. She was White, and I informed it to a counselor. I got a new class . . . [and] a different teacher, and that teacher is better for me because he actually cares of what grade I get in this class. So I'm pretty proud of that. [Lisa, Latina, female, age 14][3]

When Lisa experienced a racist remark from her teacher, she took action by communicating her experience to someone in authority (i.e., a counselor) to change into a different class with a new teacher who cared about her and her learning. By all accounts, this situation met our understanding of agentic engagement as a proactive attempt to influence classroom instruction, even though it did not look like the rest of the situations that students discussed. We highlight this example to show that students did overwhelmingly enact their agency within the classroom, but sometimes they felt that they had no other choice but to operate outside the classroom to change the instruction that they received so that it better served them.

Criteria for Agentic Engagement

Thus far, we have presented our findings that students used a variety of strategies to intentionally and constructively solicit greater support from their teachers, change their personal experience, and transform the learning environment for the classroom as a whole. However, we found that there was substantial variability in the extent to which students believed that agentic engagement could actually "work." This variability was mostly within the student: students talked about how "it depends" on certain factors, or it can(not) work because of some reason. Therefore, the third theme describes the criteria for agentic engagement, or the conditions, situations, and characteristics that students believed to be important for the agentic engagement process. These criteria were further demarcated into teacher factors and student factors.

Beginning with the teacher-related factors, the most common criteria that impacted students' beliefs about the use and effectiveness of agentic engagement strategies centered around teacher discouragement or encouragement for such behavior. More students discussed feeling discouraged by their teachers, versus feeling encouraged, to enact strategies for agentic engagement.

Students felt discouraged by their teachers through their teachers' demeanor, practices, and lack of responsiveness to students' attempts at influencing the flow of instruction. They discussed their teachers' negative or defensive reactions when they tried to make suggestions in class. For example, Tina shared, "I feel that not all teachers take that advice that you give them to change. Like, they'll take it as you're throwing shade at their teaching skills, or like, they just don't implement what you want in the class" [Tina, Black, female, age 18]. Students also discussed their teachers' controlling teaching style as a barrier to their agentic engagement. As Tanya described, "[the teacher will] tell you like,

[2] To guard against overcounting of comments, areas in the transcript where a student discussed a type of comment were only coded once. However, if the conversation shifted/changed, and then the student made another (separate and new) comment of the same type later in the discussion, both comments of that type were counted.

[3] During the session, the student first briefly mentioned this experience and then the moderator asked follow-up questions to learn more. This statement was shortened, and the order of the sentences was changed for brevity.

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'No, I did this for this many years . . . So I know it and you don't. Let me teach it and you just take in what I tell you'" [Tanya, Black, female, 17]. Teacher indifference to agentic engagement was not confrontational, but still problematic. For instance, Maria said, "My chemistry class, it's mostly book work. We've had four labs the whole year. And if we ask him if we can do a lab, he says yes, but we never do. He says, 'it's a lot of work'" [Maria, Latina, female, age 15]. Importantly, there were situations in which the teacher may not have intended to suppress students' attempts at agentic engagement, but it was still perceived as such by the students. For example, some students described how their teachers would tell them to hold their questions or suggestions until after class, but students would forget to do so. When students were asked to hold their thoughts, and neither party followed up on those at a later time, students still felt that their teachers did not care.

Teachers' hostile, defensive, or indifferent responses to students' questions, suggestions, and input resulted in students being "scared to go over to them" and a sense of powerlessness. From the students' perspectives, there was a clear link between their teachers' negative reactions to their agentic engagement and a decline or cessation in their use of agentic behaviors in the classroom. The following statement by Destiny is an example of this link:

> That's why I don't even ask [my teacher] questions anymore because if, like, sometimes she doesn't even answer my question, and sometimes she'll insult me, like, "you're about to be a senior in this many months." And I'm like, "that's not what I'm asking, can you answer my question?" Stuff like that. So it makes me just not ask her anything. [Destiny, Black, female, age 18]

Teacher discouragement was closely related to the idea that the "teacher has more power" and authority relative to the student. That is, "the teacher kind of holds all the cards so they have all the authority. You cannot just demand things to be in a certain way that's more suitable to your learning" [Isabelle, White, female, age 15]. We did not detect any patterns by race and/or gender when examining the number of students who discussed teacher discouragement as a factor that influenced their agentic engagement. However, we do note that Black girls who made teacher discouragement comments were more vocal about their frustration and dissatisfaction (36% of these comments were said by Black girls). Moreover, of the comments that students made related to their experiences or beliefs that agentic strategies do not work to solicit support from the teacher (such as the one above from Destiny), nearly half (45%) of such statements were made by Black girls.

In direct contrast to teacher discouragement, students discussed that they used agentic strategies and felt that their attempts worked when their teachers encouraged them through their teachers' demeanor, practices, and levels of responsiveness to their thoughts, ideas, and questions. Students discussed their teachers' positive responses when they enacted their agency in the classroom and how that led to better learning experiences. For example, Miguel described his teacher's enthusiastic and supportive reaction after a student made a suggestion: "You could see in her eyes, and she was like 'Yeah, we could do that.' So then she created this assignment, and we all liked it" [Miguel, Latino, male, age 17]. Students discussed the importance of teachers giving "reminders" or "reassurance" in telling their students that "this is an open environment—you can ask questions," and without this they did not think that they "would be as comfortable giving my input in the classroom" [Karina, Latina, female, age 16]. However, the more common pattern was a description of a collection of practices that created a positive classroom climate that supported their agentic engagement.

Students believed that teachers created a climate for agentic engagement by welcoming student input in a variety of ways. For example, Chris said,

> A lot of teachers always request students for feedback. They always ask, "What can I do better? What am I doing wrong?" He will give us a survey that asks how the unit was, how we felt about it, what we liked about it, and what we didn't, and he asks about how he can improve his experiments. And he just wants our feedback. He says, "Help me help you." [Chris, Latino, male, age 16]

In addition, students discussed that teachers who encourage them to agentically engage genuinely listen to students "in case they have anything to share" and are highly receptive to their suggestions and find ways to incorporate what they want to do, "or something similar to what we want to do." They also connect with them on a personal level, make them "feel safe at school," and are highly attuned to their students' emotions and behaviors.

As with teacher discouragement, students also made the direct link between the encouraging classroom climate and their successful experiences with agentic engagement. Tony's description is one example:

> I think we were talking about mechanical energy. I remembered that I had a little fun science game at home, and because I felt comfortable with the classroom, I was able to bring that to the classroom and actually show everyone—show the teacher, like—I asked him how this actually works with everything because I didn't fully understand it. So, it was good to hear someone who I trust, who knows everything about this to confirm what I thought. Just bringing new stuff that I didn't know, and I really like how it helps. [Tony, Latino, male, age 14]

Tony brought something from home to initiate a discussion with his whole class—a highly volitional agentic behavior—and he did so because of the comfortable classroom climate that his teacher had created. Indeed, "comfortable" was a word that students often used to describe a classroom climate that supported student agency. Teacher encouragement and discouragement were both closely related, and oftentimes overlapped, with the idea that the student-teacher relationship was important and could affect student use and beliefs about the effectiveness of agentic strategies. That is, "if you do not have a good relationship with [the teacher] . . . you're not gonna feel motivated to talk to them" [Tanya, Black, female, age 17]. Of note, we detected gender differences in teacher encouragement comments. Boys (across race) accounted for 69% of those who discussed teacher encouragement for agentic engagement; they also accounted for 77% of these comments.

The overarching theme of students' beliefs about teacher discouragement or encouragement for agentic engagement could be summarized as experimental: students based their use of agentic strategies and their beliefs about its effectiveness on "how [teachers] react, how they take that in." When teachers reacted

negatively, they decreased their use of agentic behaviors and believed that they could not do anything to change the instruction they received. In contrast, when teachers reacted positively, students were more likely to continue to use strategies for agentic engagement and believed that these strategies could be used to influence instruction.

We also found that students also shared concerns about their teachers as factors that could influence agentic engagement. Concerns about the teacher included not wanting to bother or upset the teacher, or not wanting to change what the teacher took time to plan for them. Similar to teacher discouragement or encouragement, students' concerns about their teachers suggests that students were actively thinking about how strategies for agentic engagement would affect their teachers if they were to make suggestions, offer input, or ask questions.

The teacher-related factors suggest that teachers play a powerful role in shaping student beliefs around agentic engagement as a viable strategy for enriching their own learning and motivation. Next, we present some student-related factors that students believed could influence the agentic engagement process.

We asked students whether agentic engagement could work for all students or only some students. Whereas five students mentioned that "everybody is capable" of agentic engagement, we found that others discussed prototypical beliefs, or beliefs about specific characteristics that might affect which types of students use (or do not use) agentic strategies, or that might affect whether agentic strategies are effective. The most common prototypical belief related to personality characteristics. For example, students believed that outgoing, extroverted, and "open" people would be more likely to use agentic strategies, compared with introverted or shy people. Other factors that were mentioned included students' levels of existing motivation, achievement and behavioral issues, student backgrounds, and gender. We provide some examples of student responses in Table 4 to show the variability in these beliefs for different types of students. Of note, we did not find any instances where students believed that the race or ethnicity of the student could influence the agentic engagement process. As Leticia explained, "I think it doesn't relate to your race. I guess, like I've mentioned before, it's all about your will. It doesn't matter if you're White, you're Black, you're purple, or pink. It's about how you motivate yourself and how you handle things yourself" [Leticia, Latina, female, age 16].

Lastly, students discussed concerns related to their fellow peers, which they believed affected whether and how they should enact their agency in the classroom. Students expressed concerns about being "embarrassed by their peers" and feeling like they would "be clowned" or judged by their classmates. For example, Alyssa said "you feel judged or like, you just think they're going to think I'm dumb, you know? So you just do not say anything" [Alyssa, Black, female, age 16]. Whereas some students expressed feelings of being concerned because of their classmates, others expressed being concerned *for* their classmates. For example, students mentioned that they should be thoughtful about how their attempts at agentic engagement might negatively impact their classmates. They said it's important to be cautious when asking the teacher to cover class content at a faster pace, "because you're not the only one in that class—there's other people who work at a different pace," and those students might suffer as a result of that request [Sergio, Latino, male, age 17].

In sum, we found that teachers and peers played a powerful role in shaping student beliefs about agentic engagement. Students understood the process of agentic engagement and described how they could enact behaviors to improve teachers' instructional practice and motivating style, enhance their own motivation and learning experience, and create a more fun and energetic learning environment for the classroom as a whole. However, this process

Table 4
Prototypical Characteristics That Students Believed Could Affect Agentic Engagement

Characteristic	Sample quotations
Personality	"I definitely think only certain kinds of students. I definitely think it would be so much harder for students who are just introverted and shy or who are just are not comfortable approaching authority and stuff" [Karina, Latina, female, 16].
Existing motivation	"I think it really works for people who are like already motivated because they're really just like 'we got it'" [Aiko, White/Asian, female, 17].
Achievement or behavioral issues	"I think [teachers] want to hear from somebody who pays more attention, at least tries and does great in class" [Tanya, Black, female, 17].
	"Say like, bad behavioral kids, they want to do something, the teacher won't believe them because they just do bad" [Yasmine, Black, female, 15].
Backgrounds	"I know a lot of people who grew up really poor and had really troubled times, and they're just—they're taught to take what you have and not really try to change it because you should be grateful for what you have. Those students might have a more difficult time trying to change something, because they'll just be like, 'this is better than what a lot of people have'. . . I think a lot of the times it's the environment that they grew up in. A lot of the times, like, parents, especially for kids like their parents are immigrants, like their parents were worse off than they have here. So their parents often teach them, 'you have more than we ever had as kids, so just be grateful for what you have'" [Miguel, Latino, male, 17].
Gender	"You have kids who can go up to teachers . . . My sister is really good at doing that–I can't do that, right. That's actually a natural disadvantage men have. That's being a little more practical because we don't have the same social skills as women do" [James, Asian, male, 16].
	"There are some stereotypical men in my classroom that are like, 'oh, women shouldn't speak up,' and [my teacher] knows to correct them. And he's like 'Don't say that in my classroom'" [Alex, Latinx, nonbinary, 14].

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was largely contingent on the learning environment, which either catalyzed or thwarted their use of agentic engagement behaviors and shaped their beliefs about the usefulness of agentic engagement for academic progress.

Discussion

The purpose of the study was to describe agentic engagement from the perspectives of students. Specifically, we focused on how urban high school students in physical science and engineering classes agentically engaged, if at all, and what beliefs they held about this process. We examined these questions through phenomenological research procedures that included interviews and focus groups with racially diverse students. Our findings revealed three overarching themes that described how, why, and when students agentically engaged in the classroom. In this section, we compare our findings with the literature on agentic engagement and discussed the implications of our study for teacher practice and future research.

Juxtaposing Findings With the Literature

First, our findings include a description of the strategies that students discussed for agentic engagement. The most common behaviors that we identified included asking the teacher questions, expressing opinions and preferences, making suggestions, and communicating interests, wants, or needs to the teacher, among other more specific behaviors. These behaviors were largely consistent with Reeve's (2013) operationalization of agentic engagement. However, in addition to Reeve's focus on student-teacher interactions, we also found that 71% of the students in our sample discussed connecting with friends or peers as ways to enhance their own learning and motivation. While Mameli and Passini's (2019) study argued for the inclusion of student-peer interactions, the particular behavior that they suggested was oppositional (i.e., stance-taking). In contrast, we found that the students in our sample were more communal–they discussed ways to connect with peers and cultivate community. We did not find any instances where students discussed defending their opinions among their peers as a form of agentic engagement.

Mameli and Passini (2019) also proposed the inclusion of original contributions in classroom discussions, and teacher opposition in terms of questioning or challenging the teacher's instructions. We did find instances in which students, more or less, introduced new issues or discussion topics. Tony's description of the time when he brought a science game from home to show everyone and initiated a discussion in class is an example, but this may have been the only clear example in our data. It could be argued that many of the behaviors that we identified could, more or less, be used to introduce new issues or discussion topics. For example, when students asked "what's the purpose of this?" we interpreted this behavior as requesting information about value, but it could also be seen as the student initiating a new topic for discussion (i.e., a discussion on the importance of the topic).

Teacher opposition in the form of students telling the teacher what they do not like, telling the teacher when they do not agree with him or her, or telling the teacher when they believed that their behavior was unfair (Mameli & Passini, 2019) was also found in our study. We interpreted these behaviors as ways that students express their needs. For example, Black girls in our sample discussed a time when they had trouble learning a topic through online activities at home, because they did not have access to a home computer. They expressed their concerns to the teacher and requested to have the topic covered during in-class lecture instead. The teacher resisted at first, but later conceded and changed instruction to a lecture format once many students performed badly on a test and more students voiced their concerns. We classified these students' behavior as expressing their needs to the teacher, and specifically requesting different modes of learning (i.e., from online to in-person lecture format).

Students also discussed a variety of strategies for agentic engagement which they believed could significantly affect the success of their initiatives. These agentic approaches or styles included avoiding disrespect, approaching as a group, and being specific and solution-focused when making suggestions or requests, among others. This is a novel contribution to the literature, as prior research has not specifically addressed how students try to be constructive and strategic when agentically engaging in the classroom. Some of these strategies were more salient among particular groups of students. For example, approaching the teacher as a group (rather than alone) was a strategy that was mentioned only by girls, which might be related to gender differences in communal orientations or interaction styles (Sczesny et al., 2019). We also found that avoiding disrespect when enacting agentic behaviors and approaching the teacher privately outside of class time (in part to avoid upsetting the teacher) were especially salient among Black and Latino students. These findings are consistent with research, which has widely documented teachers' bias in responding more negatively to the behavioral or academic issues of students of color compared with White students (Gregory et al., 2010; Okonofua & Eberhardt, 2015).

Second, our findings include a description of the rationales that students discussed for agentic engagement. These were grouped into three categories: teacher support rationales, personal experiences rationales, and communal rationales. Student beliefs that agentic engagement behaviors, such as making suggestions and offering input, could directly influence their own learning experiences and indirectly work to solicit greater support for their learning from their teachers aligned with prior correlational-longitudinal studies which demonstrated that agentic engagement does, in fact, predict these outcomes. That is, agentically engaged students do report increased perceptions of teacher autonomy support and increased motivation and academic benefits, including psychological need satisfaction, achievement, and other forms of engagement (Matos et al., 2018; Patall et al., 2019; Reeve, 2013).

The literature does describe agentic engagement as a pathway to enriching and transforming learning environments (e.g., Reeve, 2013), but the focus has been on the student-teacher interaction, and not the classroom environment as a whole. Our findings demonstrated that students believed that one students' agentic engagement could significantly affect the agentic engagement of the other students in the class; and they believed that they should agentically engage to enhance the engagement of others and create a more fun and energetic environment for everyone. We did not detect differences by race and/or gender in the proportion of students who mentioned communal rationales; however, Latino students made most of these comments. Although we initially believed that students of color might experience cultural incongruity, because of the presumed tension between communal and agentic orientations

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(e.g., Boucher et al., 2017), we found that this was not the case. On the contrary, these students enacted their agency in the classroom because they wanted to help others and get others to be more active too. This finding suggests that students seemed to resolve the tension between the two orientations by developing a communally oriented agentic approach to their involvement in class. Future research may examine the extent to which such views and motivations for engagement also occur in nonurban schools, especially among White middle-class students.

Third, our findings highlight the importance of teachers and classmates in students' beliefs about agentic engagement. The importance of teachers and classmates suggests a contradiction with the concept of agentic engagement itself, as ultimately, others play a role in defining the possibility for students to agentically engage or not. When teachers were open to students' perspectives and reacted positively to students' ideas and contributions, students were likely to continue to enact agentic behaviors and believe that these strategies were useful for their learning experiences. In contrast, when teachers were controlling, defensive, dismissive, or indifferent to students' agentic engagement attempts, students were less likely to enact agentic behaviors and they felt that their teachers did not support their learning. These findings are largely consistent with theory and research which posits that an autonomy supportive teaching style (vs. a controlling or indifferent teaching style) could enhance students' agentic engagement (Reeve & Shin, 2020) as well as students' other forms of engagement and motivational experiences (Patall & Zambrano, 2019; Reeve, 2009). Findings also aligned with the extant literature on the importance of student–teacher relationships for student engagement and achievement in school (Roorda et al., 2011). Students' perceptions of teacher power and authority and concerns about being teased or judged by their classmates were also consistent with a prior study (Pineda-Báez et al., 2019). In addition to these findings, we found that participants had prototypical beliefs about which types of students enact their agency successfully, as well as other concerns related to not wanting to negatively affect their teachers and classmates with their requests.

We did find some group differences with regard to student-teacher interactions. We found that teacher encouragement for agentic engagement was particularly salient among boys. This finding was consistent with quantitative research, which found that girls reported lower levels of perceived teacher autonomy support relative to boys in chemistry, physics, and engineering classes (Patall, Steingut, Vasquez, et al., 2018). We also found that Black girls were especially vocal about their experiences with teacher discouragement and their feelings of frustration and dissatisfaction in their attempts to solicit support from their teachers. Research on teacher bias and stereotypes has found that Black girls' expressions of contrary views are often perceived by teachers as challenging authority or being defiant or "aggressive" (Blake & Epstein, 2019; Epstein et al., 2017; Morris, 2007), which may explain Black girls' disproportionate experiences of teacher discouragement for agentic engagement in our findings.

Implications for Future Research

The findings of this study carry implications for future research on agentic engagement. Strong and consistent evidence of connecting with peers in this sample suggests an opportunity to consider exploring these behaviors as a type of agentic engagement in the classroom, and perhaps testing its inclusion in existing measurement scales or as separate subscales. Research should also consider looking beyond the student-teacher interaction to student-peer interactions, to understand whether students influence each other's agentic engagement during classroom instruction or learning activities, as well as whether these intentions predict unique variance in academic outcomes. Although almost all of the instances of agentic behaviors occurred within the context of the classroom, we did find that two students discussed exercising their agency outside of the classroom (i.e., changing their teacher) to influence the instruction they received inside of the class. As such, we suggest considering multiple units of analysis (e.g., school level, class level) when investigating students' agentic engagement.

Our findings also identified a wide range of contextual (de-)motivators that could catalyze or hinder students' use of agentic engagement and shape their beliefs about its effectiveness. In creating interventions or programs aiming to boost student agentic engagement, researchers should consider the suggestions and concerns that students expressed in this study. For example, if students do not feel comfortable approaching the teacher with their questions or requests, asking a friend to accompany them may be beneficial, particularly for girls. Similarly, students who feel shy may benefit from alternative strategies, such as leaving the teacher an anonymous note or sending them an e-mail. Consistent with prior research, programs or interventions should also consider appealing to students' communal motivations for helping others in encouraging them to agentically engage (e.g., "Sharing what you do or do not like will help the teacher better teach you, and it might help your classmates learn better, too"). Though prior research suggests that students from all backgrounds benefit from prosocial or communal rationales (Brown et al., 2015); some research suggests that these may be especially impactful for Latino, Black, and Native/Indigenous groups of people (e.g., Zambrano et al., 2020).

The question of what leads students to not implement agentic engagement while believing and stating its positive value was partially explained in this study. Students expressed the benefits of agentic engagement (see Rationales theme), but they also noted a variety of factors that interfered with (or facilitated) implementation in the classroom (see Criteria theme), such as the nature of the student-teacher relationship, student-teacher power dynamics, feeling shy, feeling judged or teased by their peers, and believing that teachers only want to hear from students that perform well in the class, among other factors. Future research should attempt further understanding of these (and potentially other) hindrances to agentic engagement, including how they manifest and how they affect student implementation of agentic engagement strategies in real life. Along these lines, our study found that students are regularly trying to learn in controlling or indifferent classroom environments, which suppresses students' agentic engagement and limits their academic potential. Researchers should continue to study best practices to promote autonomy supportive teaching styles, and also encourage the application of these practices to the real world. In doing so, education researchers should consider gender, race, and their intersection, particularly when working with minoritized populations in science and engineering contexts.

The primary aim of the present study was to understand agentic engagement from students' perspectives, and with the use of the

broad methodological lens of phenomenology, our findings include a comprehensive description of the phenomenon. The trade-off with our broad aim and lens, however, is that we were limited in our analysis of potential narratives and discourses related to agentic engagement, gender, and race. As such, future research studies could apply narrative or discursive research methods for a deeper and more nuanced analysis of agentic engagement and the lived experiences of diverse learners.

Implications for Teacher Practice

This study contributes to knowledge about what agentic engagement looks like in real science classrooms from students' perspectives. As such, this study has implications for aiding teachers in recognizing agentic behaviors when they occur in their own classrooms and understanding the implications of those behaviors. An important first step to promoting agentic engagement is to understand that agentic behaviors, such as making suggestions, expressing preferences, and offering input, are usually not intended to offend the teacher. Our findings indicated that students actively thought about how to frame their concerns in a respectful and constructive way to recruit teacher support and promote their learning and interest in class topics and activities. However, students can and do experience negative emotions, frustration, and dissatisfaction in response to teacher instruction and class activities. When this happens, we believe teachers can promote mutual understanding by acknowledging and accepting students' expressions of negative affect, displaying patience, and reflecting on their own emotions and interpretations of student concerns (for example, did I get defensive when students expressed that they didn't like today's lab activity? Were students trying to insult me as a teacher? Or were they giving me valuable feedback to help me improve the way I'm teaching this topic so that they could engage and learn better?). Moreover, our findings suggest that how teachers encourage and respond students' contributions matters. Teachers can support students' agentic engagement by adopting an autonomy supportive teaching style. Broadly, this teaching style includes an inviting tone, asking students what they want, listening with an open mind to what they say, being responsive to their initiatives, and appreciating their contributions and efforts in helping to improve the learning environment (Reeve, 2009). For more information on autonomy supportive teaching practices, we recommend Reeve and Shin (2020) and Patall and Zambrano (2019); both are written for practitioner audiences.

Conclusion

The present study aimed to conduct a deep investigation of high school students' beliefs about agentic engagement, which could not be sufficiently examined through quantitative or observational methods alone. Results revealed that agentic engagement is a meaningful construct to urban high school science students and is understood to have the potential to reshape the learning experience, provided a variety of important conditions are met. We hope researchers and educators can use the results of this study to continue to develop understanding of agentic engagement and leverage its potential for creating recursive and sustaining cycles of motivation in the classroom.

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Appendix A

Interview/Focus Group Protocol

Welcome everyone. My name is _____ and this is _____. We are both from the University of Southern California and affiliated with the Motivation and Education Research Group led by Dr. Erika Patall.

Thank you for coming to talk with us today about your motivation and engagement in physical science class. The purpose of these focus groups is to try to identify strategies that will aid in engagement and motivation in physical science classes. We are interested in your honest opinions and feedback so there are no right or wrong answers. You are the experts on your motivation in science class, so we want to know what you really think. You do not need to agree with others, but you must listen respectfully as others share their views. One other rule, please silence your cell phones while we're discussing and keep them out of view.

My role as moderator is just to guide your discussion and my assistant moderator will help me with that by taking notes. But our goal here is for you to have a discussion and talk to each other. Feel free to respectfully comment on ideas other participants present. You've probably noticed the recorder and camera. We're tape recording the session because we don't want to miss any of your comments. People often say very helpful things in these discussions, and we can't write fast enough to get them all down. Especially because we are recording, please only one person speaks at a time.

We will be on a first name basis tonight, and we won't use any names in our reports. You may be assured of complete confidentiality. We've given you name tags to help us remember each other's names. Let's begin by going around the table. Tell us your name, your grade, and what science class you are taking.

(*Appendices continue*)

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Great, thank you. Now, the first thing I'd like to do is give you a figure to describe what we think is happening when students are feeling motivated in a physical science class. Take a minute to look at the figure and make any notes, changes, additions, edits, and the like to it.

SEE FIGURE

(Warm-up question)

1. Does anyone want to say what they think the figure is trying to describe?

   a. Do you like this figure and why or why not?

   b. Possible probes:

      i. Would you explain further?

      ii. Would you give an example?

      iii. I don't understand.

(Use the description someone gives to provide a description of what the model is attempting to describe.)
Thanks for that description! So, the way we think about what is happening in the classroom and why students either feel interested, engaged, and motivated in class, or do not, in part starts with the teacher. Teachers find ways to create activities that connect to things that students are interested in, they can be very caring toward students, they can help students see why what they are learning is important or useful to them, they can be understanding of students' perspectives and even their negative feelings in class, among many other practices. And this all helps students feel motivated and engaged in science class and feel like they want to learn. There are several ways that students feel motivated and engaged. Those include being engaged in terms of your behavior or the way you act, your emotions or the way you feel, and your cognitions or thinking. But you'll notice we have really emphasized this one type of engagement to the side and also have some arrows that go from students' engagement back to what teachers' do. What that is about is that we think that students play a powerful role in shaping their own motivation and helping teachers support their motivation. The idea is that by being authentic and expressing your preferences, suggestions, and asking questions, even at times when you do not feel all that motivated in class, you can change how you feel and help the teacher find ways to support your motivation.

(Primary questions)

2. What do you think of this idea that students can change their own motivation in physical science class with the way they engage in class?
   (probing questions)

   a. Why do you think it works (OR doesn't work) like this?

   b. Why do you think that?

   c. Can you give me an example?

   d. Would you explain further?

   e. I don't understand.

3. What about the idea that students can improve how teachers' support students' motivation in physical science class with the way they engage in class?

   (probing questions)

   a. Why do you think it works (OR doesn't work) like this?

   b. Why do you think that?

   c. Can you give me an example?

   d. Would you explain further?

   e. I don't understand.

4. Now thinking about the things students could do during class to change how they feel or help the teacher make class more motivating, what are some strategies that students can use to help create a motivating physical science class, even when you are having a hard time being engaged?

(*Appendices continue*)

(probing questions)

a. Do you have examples of those working for you or someone else?

b. Who will that work for? Do you think that could work for all students or students from all kinds of backgrounds?

c. Why (OR Why not)?

d. How do teachers react when students do that?

e. Are there things students really should not do?

f. Why (OR Why not)?

g. Would you explain further?

h. I don't understand.

5. Of everything you have heard today, what do you think would be the most helpful for other students to understand to help with their motivation in science class? Summarize the suggestions (or ask assistant moderator to summarize) what you heard participants give and ask if you missed anything.

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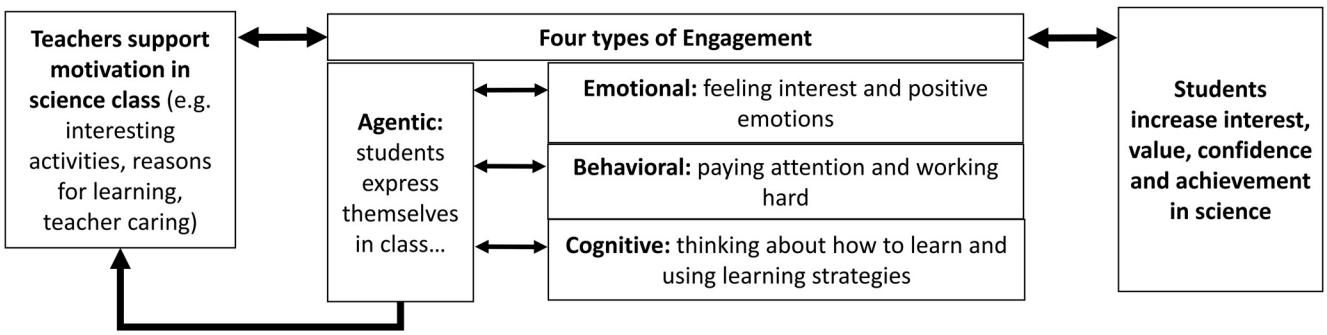
Appendix B

Figure Presented to Participants During Interview/Focus Group Session



Note. Edits were made to help with comprehension for adolescent participants. From "How Students Create Motivationally Supportive Learning Environments for Themselves: The Concept of Agentic Engagement," by J. Reeve, 2013, *Journal of Educational Psychology*, 105, p. 580. Copyright 2013 by the American Psychological Association. Adapted with permission.

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